and content trends.

Task 4 – Trend Analysis by Release Year

[3] 0s

```
# Organize content by release year
yearly_content = df.groupby(['release_year', 'type']).size().unstack(fill_value=0)

# Display the result
yearly_content.head(10)
yearly_content = cleaned_df.groupby(
    ['release_year', 'type']
).size().unstack(fill_value=0)

yearly_content.head(10)
```

```
---------------------------------------------------------------------------
NameError                                 Traceback (most recent call last)
/tmp/ipykernel_1274/1020100296.py in <cell line: 0>()
      1 # Organize content by release year
----> 2 yearly_content = df.groupby(['release_year', 'type']).size().unstack(fill_value=0)
      3
      4 # Display the result
      5 yearly_content.head(10)

NameError: name 'df' is not defined
```

Next steps: Explain error

[ ] 0s

```
# Calculate total content released each year
yearly_total = cleaned_df['release_year'].value_counts().sort_index()
```

NetflixDataset_Analysis.ipynb

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RAM Disk

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[28] ✓ 0s

```
# Load the original Netflix dataset
df = pd.read_csv('/content/Dataset.csv')


print("Dataset loaded successfully!")
print("Number of rows:", df.shape[0])
print("Number of columns:", df.shape[1])
```

```
Dataset loaded successfully!
Number of rows: 8790
Number of columns: 10
```

[29] ✓ 0s

```
# Display the first 5 records
df.head()
```

	show_id	type	title	director	country	date_added	release_year	rating	duration	listed_in
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	United States	9/25/2021	2020	PG-13	90 min	Documentaries
1	s3	TV Show	Ganglands	Julien Leclercq	France	9/24/2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...
2	s6	TV Show	Midnight Mass	Mike Flanagan	United States	9/24/2021	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries
3	s14	Movie	Confessions of an Invisible Girl	Bruno Garotti	Brazil	9/22/2021	2021	TV-PG	91 min	Children & Family Movies, Comedies
4	s8	Movie	Sankofa	Haile Gerima	United States	9/24/2021	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies

Next steps: Generate code with df New interactive sheet

Variables Terminal

✓ 13:48 Python 3

[30] ✓ 0s

```
# Display column names and data types
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 8790 entries, 0 to 8789
Data columns (total 10 columns):
 #   Column        Non-Null Count  Dtype 
---  ------        --------------  ----- 
 0   show_id       8790 non-null   object
 1   type          8790 non-null   object
 2   title         8790 non-null   object
 3   director      8790 non-null   object
 4   country       8790 non-null   object
 5   date_added    8790 non-null   object
 6   release_year  8790 non-null   int64 
 7   rating        8790 non-null   object
 8   duration      8790 non-null   object
 9   listed_in     8790 non-null   object
dtypes: int64(1), object(9)
memory usage: 686.8+ KB
```

[31] ✓ 0s

```
print("Column Names:")
print(df.columns.tolist())
```

```
Column Names:
['show_id', 'type', 'title', 'director', 'country', 'date_added', 'release_year', 'rating', 'duration', 'listed_in']
```

[32] ✓ 0s

```
# Count missing values in each column
missing_values = df.isnull().sum()

print("Missing Values:")
print(missing_values)
```

```
Missing Values:
```

[32] ✓ 0s

```
missing_values = df.isnull().sum()

print("Missing Values:")
print(missing_values)
```

```
Missing Values:
show_id         0
type            0
title           0
director        0
country         0
date_added      0
release_year    0
rating          0
duration        0
listed_in       0
dtype: int64
```

[33] ✓ 0s

```
missing_percentage = (df.isnull().sum() / len(df)) * 100

missing_summary = pd.DataFrame({
    'Missing Values': df.isnull().sum(),
    'Percentage': missing_percentage.round(2)
})

missing_summary
```

	Missing Values	Percentage
show_id	0	0.0
type	0	0.0
title	0	0.0
director	0	0.0

NetflixDataset_Analysis.ipynb

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RAM Disk

[44] ✓ 0s

```
cleaned_df = pd.read_csv('/content/Netflix_Cleaned.csv')

print("Cleaned dataset loaded successfully!")
print("Rows:", cleaned_df.shape[0])
print("Columns:", cleaned_df.shape[1])
```

```
Cleaned dataset loaded successfully!
Rows: 8790
Columns: 10
```

[45] ✓ 0s

```
# Count Movies and TV Shows

content_counts = cleaned_df['type'].value_counts()

print("Number of Movies:", content_counts.get('Movie', 0))
print("Number of TV Shows:", content_counts.get('TV Show', 0))
```

```
Number of Movies: 6126
Number of TV Shows: 0
```

[46] ✓ 0s

```
total_content = content_counts.sum()

print("Total Netflix Content:", total_content)
```

```
Total Netflix Content: 8790
```

[47] ✓ 0s

```
content_percentage = (content_counts / total_content) * 100

print("Content Proportion:")
print(content_percentage.round(2))
```

```
Content Proportion:
type
```

Variables Terminal

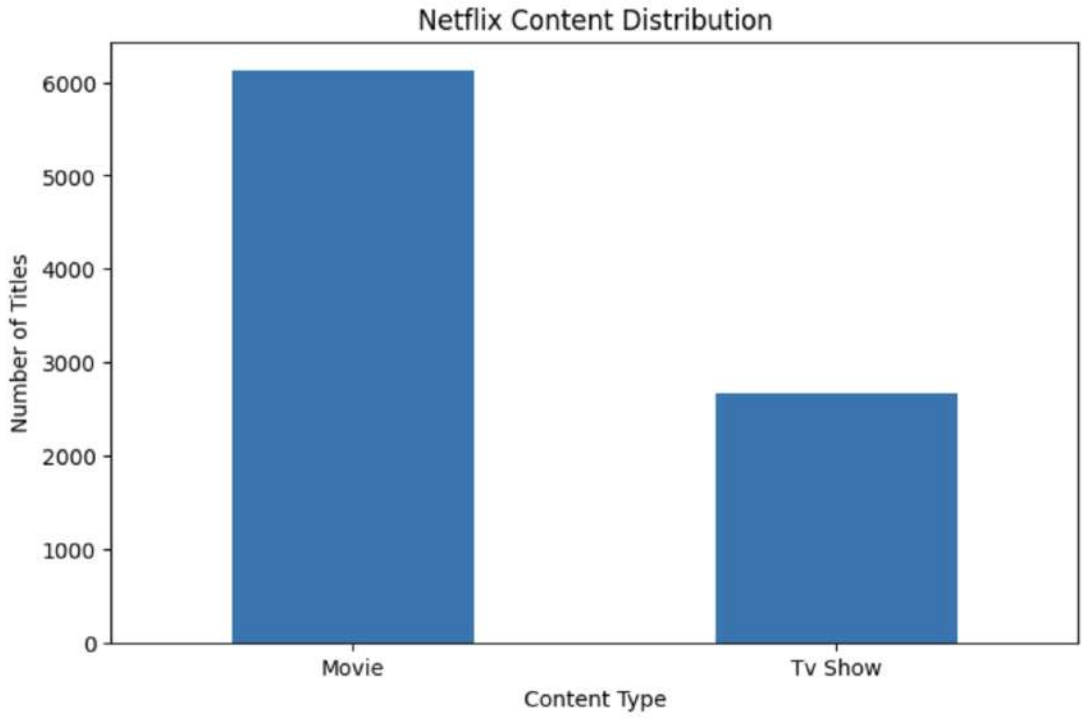
✓ 13:48 Python 3

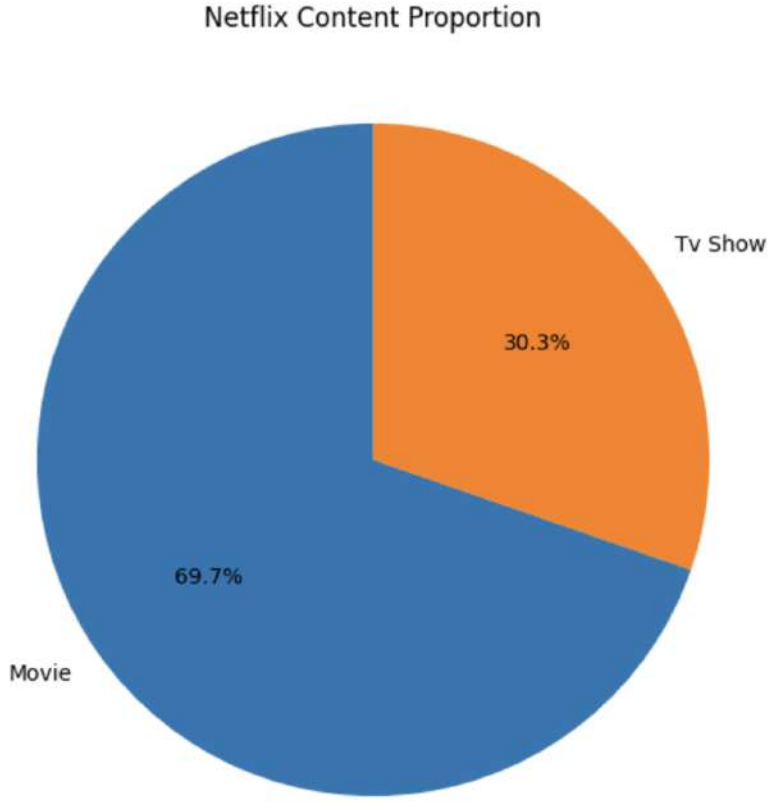
[48] ✓ 0s

```
content_counts.plot(kind='bar')

plt.title('Netflix Content Distribution')
plt.xlabel('Content Type')
plt.ylabel('Number of Titles')
plt.xticks(rotation=0)

plt.show()
```





[50]
✓ 0s

```
movie_count = content_counts.get('Movie', 0)
tv_show_count = content_counts.get('TV Show', 0)
```

[54] ✓ 0s

```
cleaned_df = df.copy()

cleaned_df.head()
```

	show_id	type	title	director	country	date_added	release_year	rating	duration	listed_in
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	United States	9/25/2021	2020	PG-13	90 min	Documentaries
1	s3	Tv Show	Ganglands	Julien Leclercq	France	9/24/2021	2021	TV-MA	1 Season	Crime TV Shows, International TV Shows, TV Act...
2	s6	Tv Show	Midnight Mass	Mike Flanagan	United States	9/24/2021	2021	TV-MA	1 Season	TV Dramas, TV Horror, TV Mysteries
3	s14	Movie	Confessions of an Invisible Girl	Bruno Garotti	Brazil	9/22/2021	2021	TV-PG	91 min	Children & Family Movies, Comedies
4	s8	Movie	Sankofa	Haile Gerima	United States	9/24/2021	1993	TV-MA	125 min	Dramas, Independent Movies, International Movies

Next steps: Generate code with cleaned_df | New interactive sheet

[55] ✓ 0s

```
import pandas as pd

df = pd.read_csv('Dataset.csv')

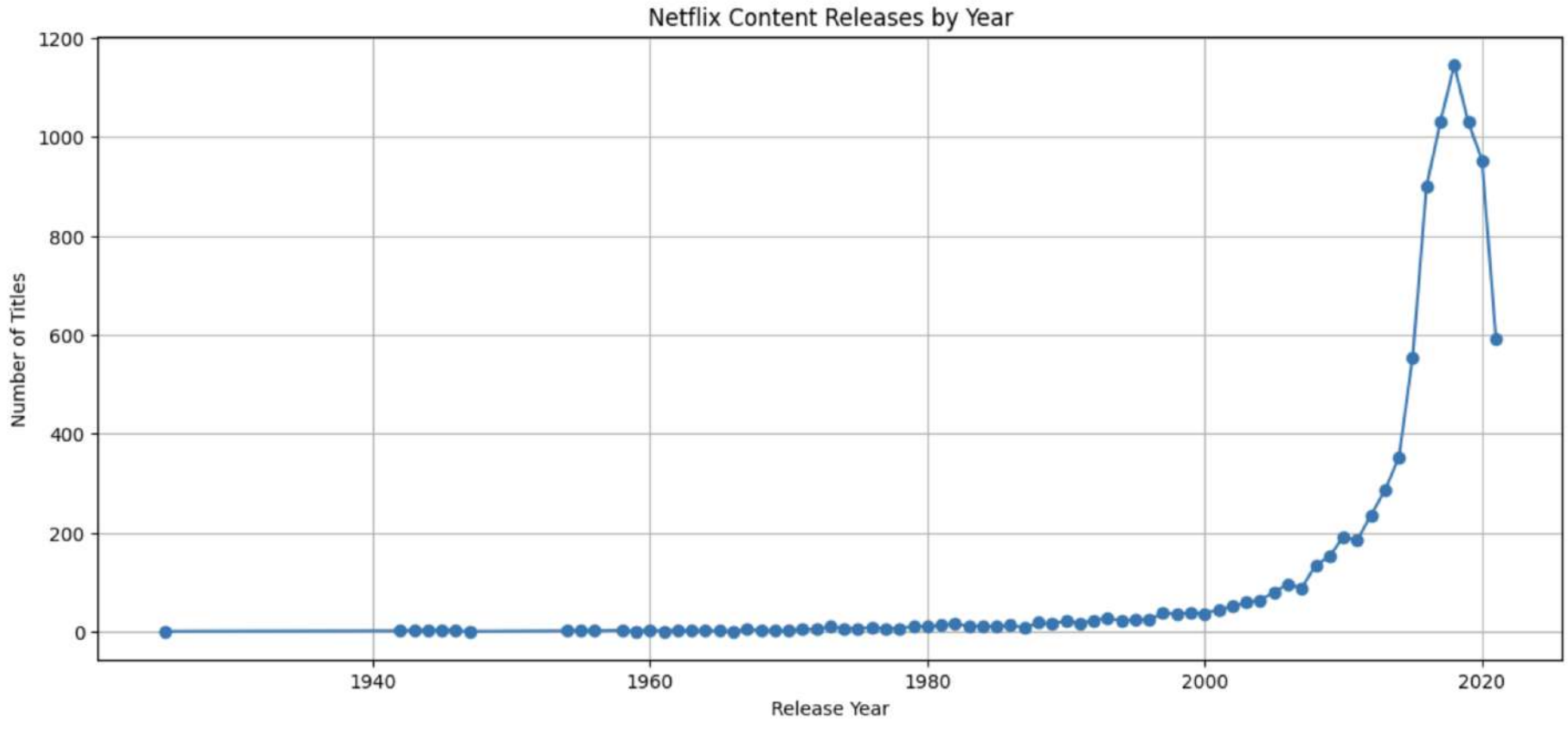
df.head()
```

	show_id	type	title	director	country	date_added	release_year	rating	duration	listed_in
0	s1	Movie	Dick Johnson Is Dead	Kirsten Johnson	United States	9/25/2021	2020	PG-13	90 min	Documentaries
		TV		Julien						Crime TV Shows, International TV Shows

[60] ✓ 0s

```
plt.grid(True)

plt.show()
```



NetflixDataset_Analysis.ipynb

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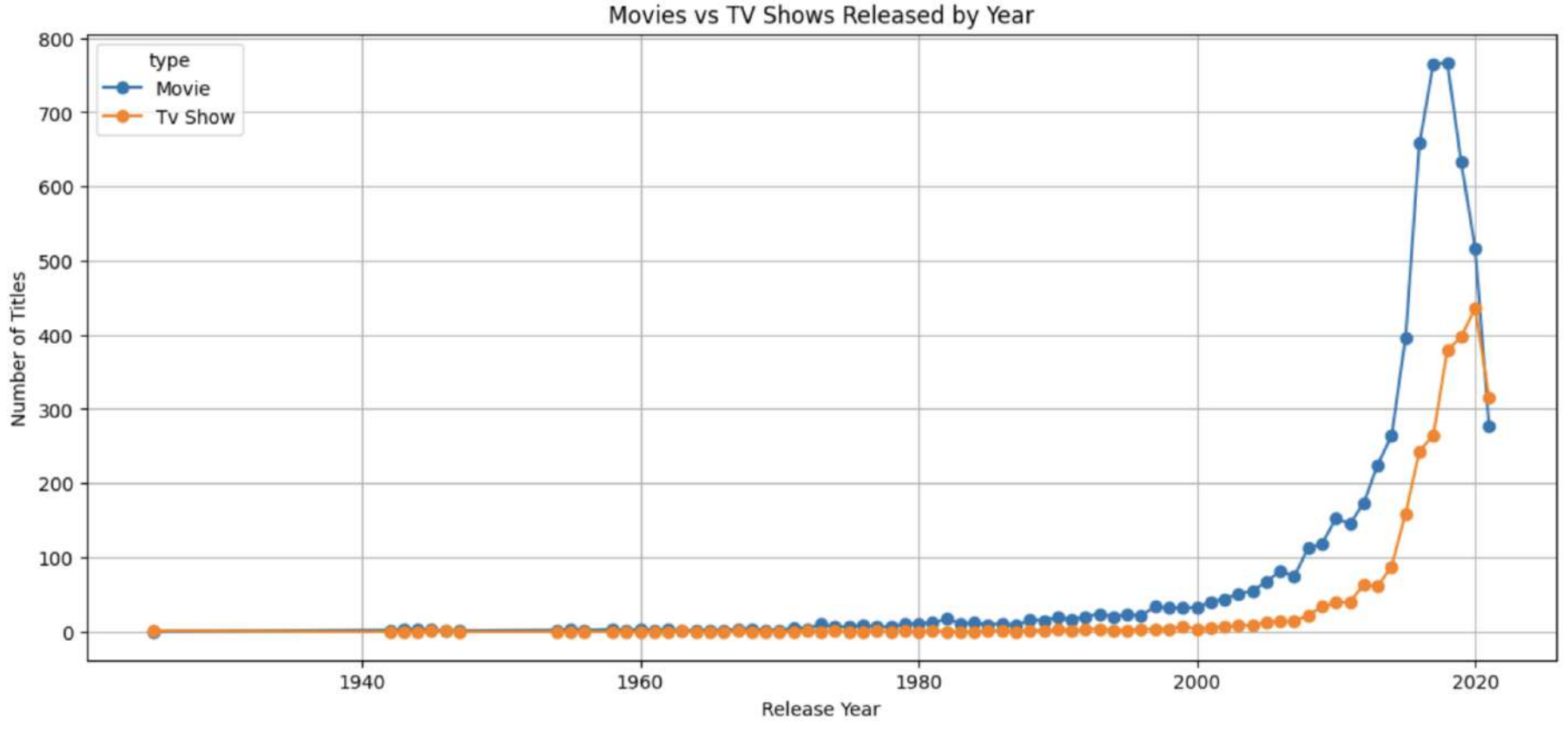
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[61] ✓ 0s

```
plt.show()
```

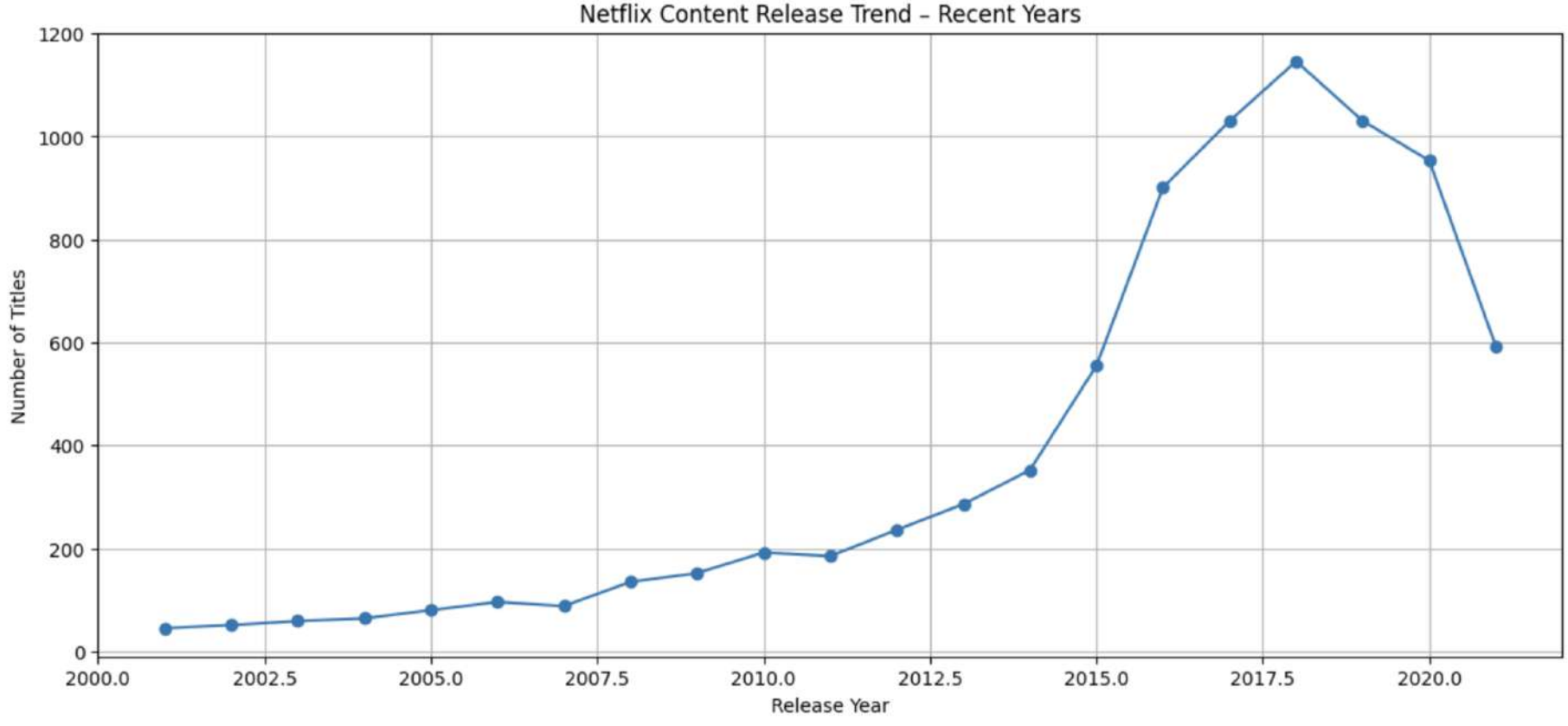


[62]

```
recent_years = yearly_releases[
```

Variables Terminal

✓ 13:48 Python 3

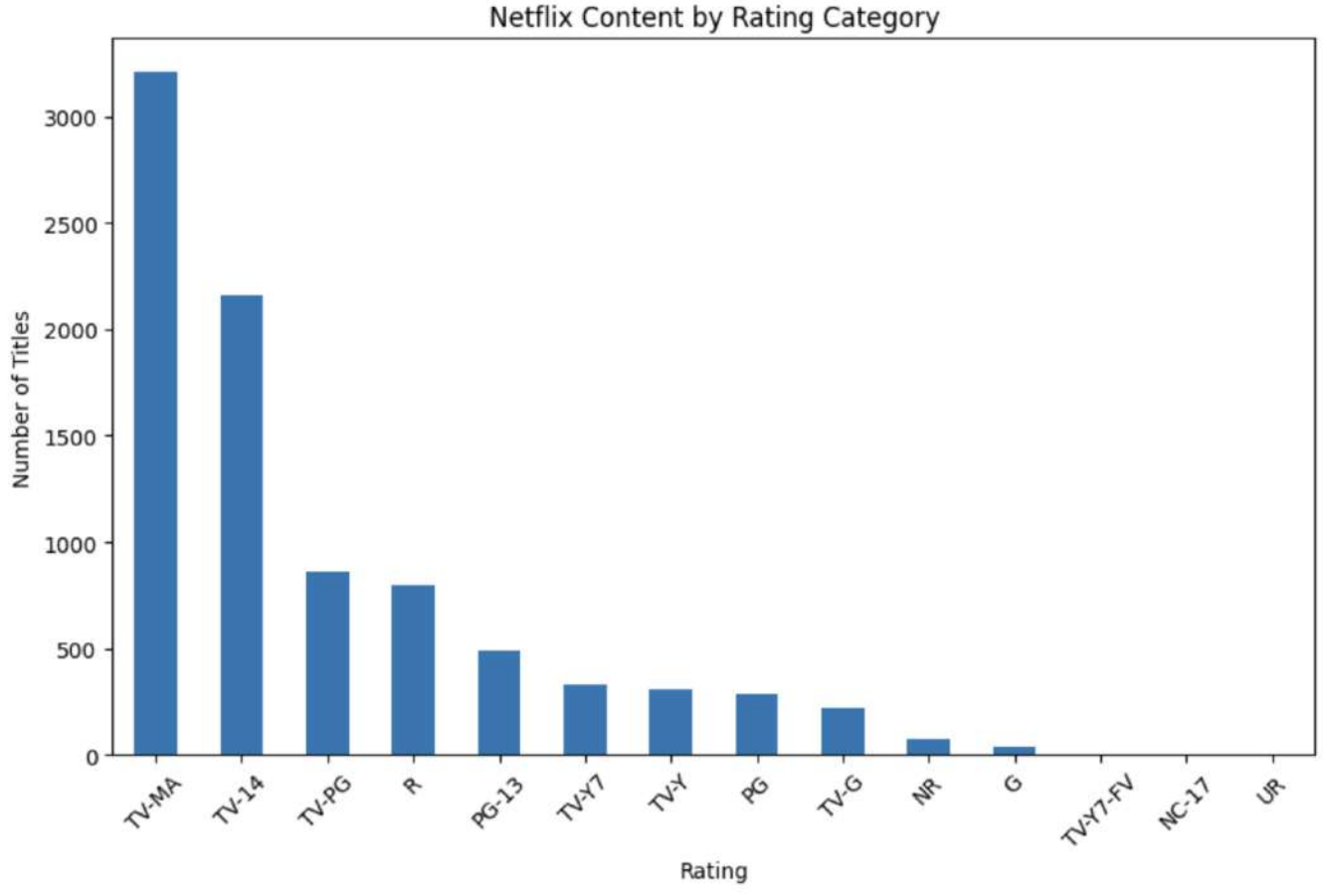


Task 5 – Content Rating & Genre Analysis

+ Code + Text

[63]

```
plt.show()
```


Netflix Content by Rating Category
Number of Titles
0
500
1000
1500
2000
2500
3000
TV-MA
TV-14
TV-PG
R
PG-13
TV-Y7
TV-Y
PG
TV-G
NR
G
TV-Y7-FV
NC-17
UR
Rating

NetflixDataset_Analysis.ipynb

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RAM Disk

[65] ✓ 0s

```
# Step 2: Identify the most common genres

genre_data = cleaned_df['listed_in'].dropna().str.split(', ').explode()

genre_counts = genre_data.value_counts()

print("Top 15 Genres:")
print(genre_counts.head(15))
```

```
Top 15 Genres:
listed_in
International Movies        2752
Dramas                      2426
Comedies                    1674
International TV Shows      1349
Documentaries                869
Action & Adventure           859
TV Dramas                    762
Independent Movies           756
Children & Family Movies     641
Romantic Movies              616
Thrillers                    577
TV Comedies                  573
Crime TV Shows               469
Kids' TV                     448
Docuseries                   394
Name: count, dtype: int64
```

[66] ✓ 0s

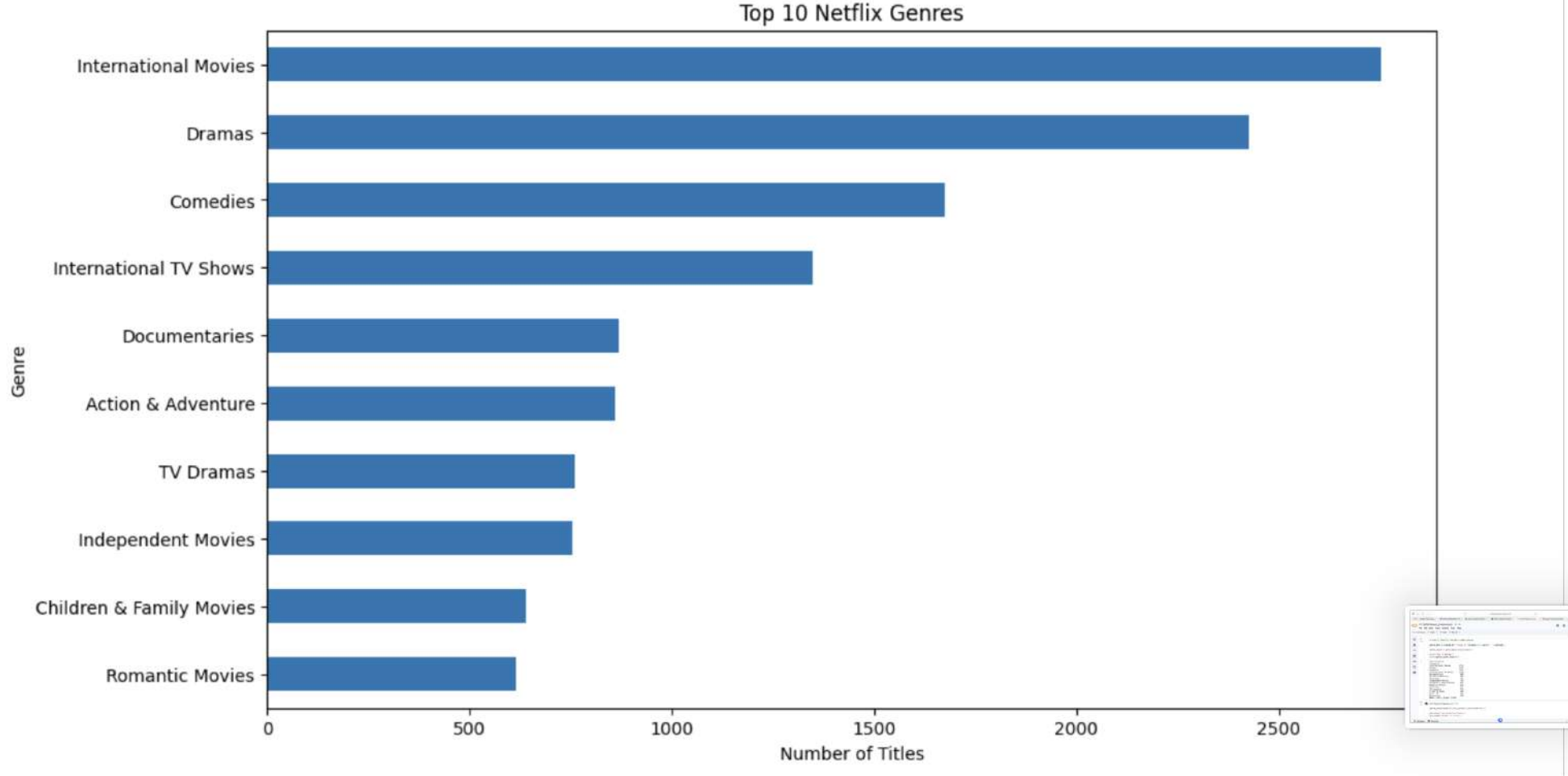
```
plt.figure(figsize=(12, 7))

genre_counts.head(10).sort_values().plot(kind='barh')

plt.title('Top 10 Netflix Genres')
plt.xlabel('Number of Titles')
```

Variables Terminal

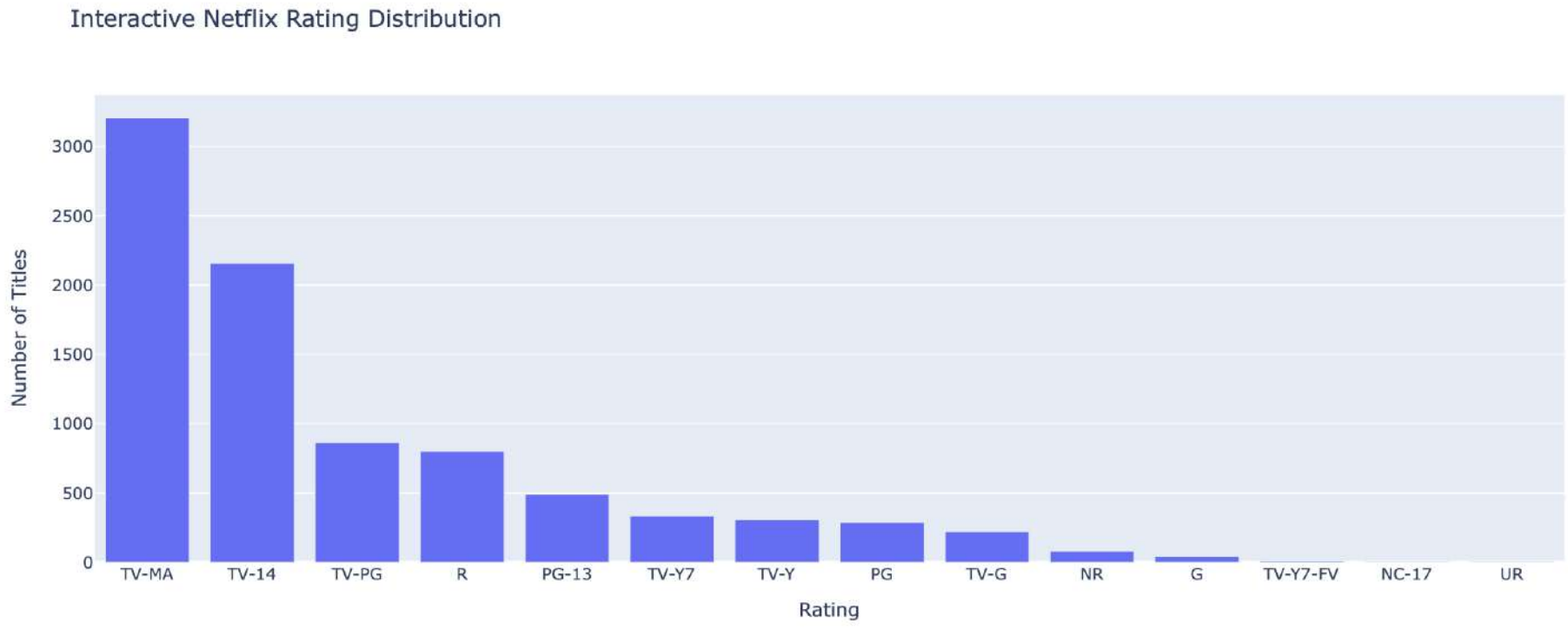
13:48 Python 3


Top 10 Netflix Genres
International Movies
Dramas
Comedies
International TV Shows
Documentaries
Action & Adventure
TV Dramas
Independent Movies
Children & Family Movies
Romantic Movies
Genre
0
500
1000
1500
2000
2500
Number of Titles

[71] ✓ 3s

```
)

fig.show()
```



[72]

```
top_genres = genre_counts.head(10).sort_values()
```

NetflixDataset_Analysis.ipynb

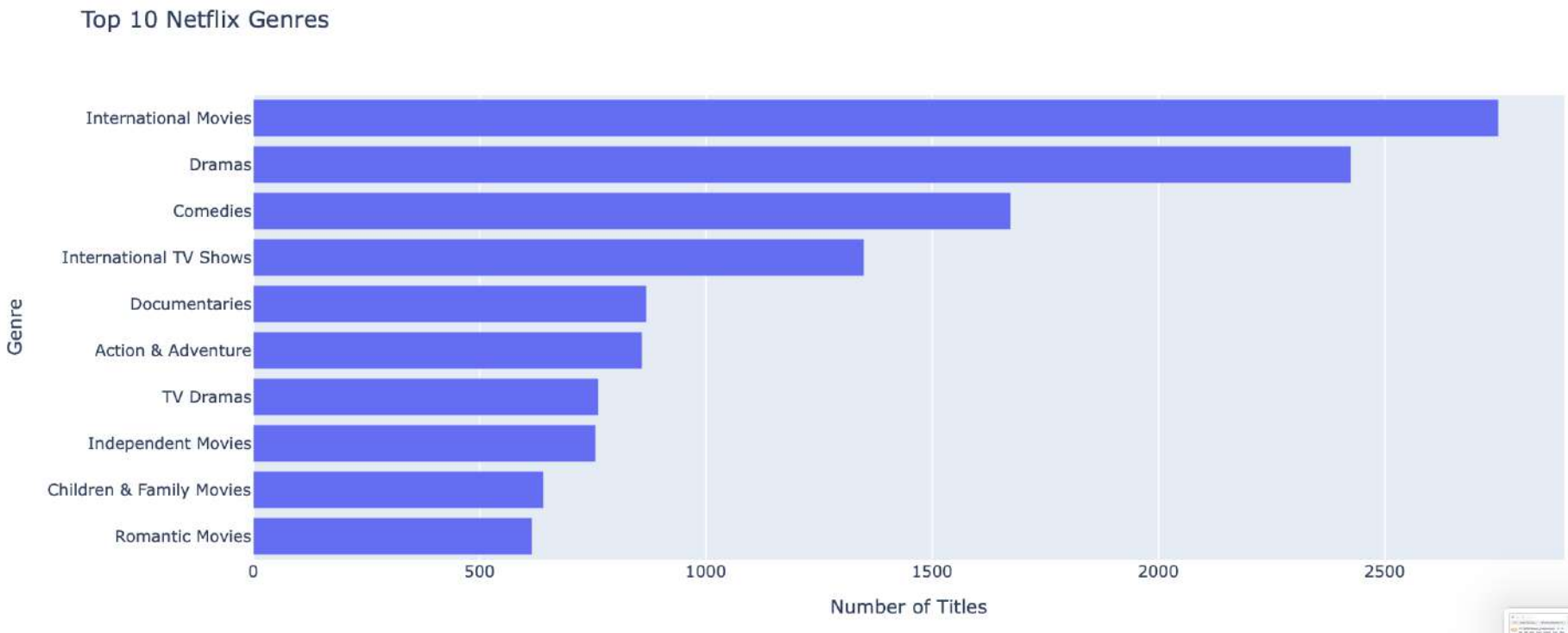
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0s



[73]

0s

```
rating_type_data = (
    cleaned_df
    .groupby(['rating', 'type'])
```

Variables Terminal

13:48 Python 3


Interactive Ratings by Content Type
Number of Titles
2000
1500
1000
500
0
G
NC-17
NR
PG
PG-13
R
TV-14
TV-G
TV-MA
TV-PG
TV-Y
TV-Y7
TV-Y7-FV
UR
Rating
Content Type
Movie
Tv Show